

9 Electricity

AS · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Electric current is charge flow rate $I = \Delta Q / \Delta t$. Potential difference is energy transferred per unit charge, $V = W / Q$. Resistance $R = V / I$; an ohmic conductor at constant temperature has a linear I-V graph through the origin. Electrical power $P = VI = I^2R$. Resistivity depends on material and temperature.

Worked example: follow the reasoning

A $6\ \Omega$ resistor carries 2 A: $V = IR = 12\ \text{V}$ and $P = I^2R = 24\ \text{W}$. Charge transferred in 5 s is 10 C.

Quick check · multiple choice

A For a resistor at fixed conditions, which expression gives power? [1]

- A I^2R
- B I/R
- C R/V
- D Q/t^2

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 A wire of length 4.0 m and area $2.0 \times 10^{-6}\ \text{m}^2$ has resistance $0.10\ \Omega$. Determine resistivity. [2]

Final answer: _____

Answers and explanation

Quick check: A. I^2R

Question 1 [2 marks]

$$\rho = RA/L [1] = 0.10 \times 2.0 \times 10^{-6}/4.0 = 5.0 \times 10^{-8} \Omega \text{ m} [1].$$

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Physics 9702, 2025–2027. <https://www.cambridgeinternational.org/Images/664565-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.